

Boost replay — fast-carb confirm fast-path

Method 2 (data-driven replay) of arXiv 2606.13882v1, on 30d of NS history across 5 users (anonymised).
Generated by `replay.py`.

What this validates

V5's observe→confirm latency commits ~1 cycle late on fast carbs (the 2026-06-16 meal: dribbled 0.3U in OBSERVING at BG 120 while V1 would've dosed 1.7U; committed late → peak 185 then crash 54). The fix: promote OBSERVING→CONFIRMED in a single cycle when acceleration is extreme. The risk is false fires on compression / exercise. This replays candidate rules `delta ≥ Dt AND deltaAccl ≥ At` over real history and scores each firing as a real meal caught earlier vs a false positive.

Candidate threshold sweep (pooled)

candidate	meals caught earlier	median lead	false fires	false rate	sleep fires
loose $\Delta \geq 8$, $accl \geq 15$	211/525 (40%)	15 min	308	2.05/day	9
mid $\Delta \geq 10$, $accl \geq 20$	97/525 (18%)	15 min	141	0.94/day	4
tight $\Delta \geq 12$, $accl \geq 30$	33/525 (6%)	15 min	53	0.35/day	3
loose+score+awake $\Delta \geq 8$, $accl \geq 15$, $score \geq 0.6$, awake	172/525 (33%)	15 min	148	0.99/day	0
mid+score+awake $\Delta \geq 10$, $accl \geq 20$, $score \geq 0.55$, awake	95/525 (18%)	15 min	108	0.72/day	0

"meals caught earlier" = of the real CONFIRMED meals, how many a fast-path fire preceded within 4 cycles (so they'd commit sooner). "lead" = how much earlier. "false fires" = fires with no real meal in the next 4 cycles (would commit on a non-meal). "sleep fires" should be ~0 (compression risk). Pick the threshold that maximises lead + catch while keeping false/sleep fires low.

Reading it / next step

Looser thresholds catch more meals earlier but fire more falsely; tighter ones are safer but slower. The build decision: choose the knee of this tradeoff, implement it in `MealHypothesis.step()` (OBSERVING→CONFIRMED gate), then re-run this replay + the shadow-equivalence report to confirm. This is decision-gate replay; full glucose-outcome Parkes needs a physiologic model (Method 3).